

Algorithm design and specific applications of artificial intelligence in the field of surface structure communications

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ABSTRACT

In the past decade, programmable metasurfaces have attracted widespread attention due to their ability to manipulate electromagnetic waves at the subwavelength scale in space. By enabling controllable manipulation of the electromagnetic propagation environment, they not only provide new means for beam steering, coverage enhancement, and spectrum reuse in communication systems, but also demonstrate tremendous application potential in fields such as optical remote sensing, holographic imaging, and antenna integration. However, the design of metasurfaces involves a vast parameter space, high-complexity electromagnetic solutions, and nonlinear spatial coupling. These challenges make traditional electromagnetic simulation and optimization methods inefficient and difficult to achieve real-time control. Meanwhile, artificial intelligence, especially methods such as machine learning, deep neural networks, generative models, and reinforcement learning, has provided new ideas for handling complex nonlinear scenarios. This paper aims to comprehensively review the applications of these artificial Intelligence (AI) methods in metasurface structure design and reconfigurable intelligent surface (RIS) communication control, presenting rich literature cases to illustrate how various AI algorithms drive progress in this interdisciplinary field and providing the authors' personal analysis and suggestions on the basis of this summary.

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I. INTRODUCTION

Metasurface structures, as a type of “artificially engineered” planar device, possess unique optical and electromagnetic properties and hold tremendous application potential in many fields. By altering material properties and metasurface, it is possible to flexibly manipulate the amplitude, phase, polarization, and frequency of light waves, thereby producing a variety of desired electromagnetic effects. Over the past decade, with the continuous deepening of research on metamaterials, these artificially designed structures based on effective medium theory have been widely applied to control electromagnetic waves^{1–5} and have further given rise to a number of novel physical applications, such as negative refraction,^{6–9} metalenses,^{10,11} metasurface devices,^{12–14} and high-resolution imaging.^{15,16} These achievements have not only opened up new directions in metamaterial research but also generated

strong expectations within the scientific community regarding their future applications.

As a special two-dimensional form of metamaterials, metasurfaces offer distinct advantages compared with three-dimensional structures. They are easier to design, less costly to fabricate, and at the same time exhibit low profile and low loss characteristics. As a result, metasurfaces have been widely applied in many fields, including holographic projection,^{17–22} microwaves,^{23–26} the terahertz band,^{27–29} optics,^{30,31} and electromagnetic wave manipulation of various kinds.^{32,33} In optical information processing, metasurfaces can bring breakthroughs to virtual reality devices and artificial intelligence hardware. In future 6G communications, intelligent metasurfaces may be employed for beam control, mode design, and spectrum reuse. However, the traditional design of metasurfaces is not straightforward. In the past, it largely relied on numerical

calculations and parameter optimization, requiring repeated full-wave electromagnetic simulations. When dealing with nanostructures that demand specific electromagnetic responses, the problem of inverse design often arises. Designers had to rely on intuition to predict and invert simple optical responses, which has significantly restricted the further development of metasurfaces.

Moreover, the design of metasurfaces involves a vast parameter space, high-complexity electromagnetic solutions, and nonlinear spatial coupling. These challenges make traditional electromagnetic simulation and optimization methods inefficient and unable to achieve real-time control.^{34–105} At the same time, artificial intelligence—especially methods such as machine learning, deep neural networks, generative models, and reinforcement learning—offers new approaches for addressing complex nonlinear scenarios.^{106–174}

II. AI FOR PHYSICAL DESIGN OF METASTRUCTURES

In the early stages of metasurface research, the design and optimization of unit structures mainly relied on full-wave electromagnetic simulations and parameter sweeps. Although this method is intuitive, as the parameter dimensions increase and functions become more complex, the computational cost grows exponentially, which severely limits the efficiency of multi-objective, multi-band, and multi-polarization designs. The introduction of artificial intelligence methods, especially neural network-based fitting and inverse design, has enabled researchers to quickly obtain feasible solutions within the vast design space, thereby significantly improving both the efficiency and innovativeness of metasurface physical design.

At the beginning of this evolutionary path, researchers mainly relied on forward surrogate models to assist inverse design. The core idea of such methods is to train a deep neural network capable of rapidly predicting the electromagnetic response under given geometric parameters, thereby replacing costly full-wave simulations. Subsequently, by performing exhaustive search or optimization in the parameter space, the inverse mapping from target electromagnetic responses to structures can be achieved. The most representative work was presented by Liu *et al.* in 2018,¹⁷⁷ where they demonstrated how convolutional neural networks (CNNs) can be used to predict the transmission and reflection spectra of metasurface unit cells, thus significantly accelerating the design iteration process. Similarly, in 2022, Gu *et al.*¹⁷⁵ made a breakthrough by applying multilayer perceptron (MLP) models to learn the mapping between metasurface operating parameters and imaging performance: by training the network to predict how different metasurface coding configurations affect imaging system outputs (such as image quality and resolution), they achieved rapid prediction and design feedback. Through this AI-driven inverse process, the system could directly backtrack from the target imaging performance to the required metasurface coding pattern. This represents a classic application of AI algorithms in inverse metasurface design: using MLPs to efficiently approximate complex electromagnetic relationships and guide the coding design process, thereby making design more intelligent and efficient while avoiding the high cost and cumbersome operations of traditional simulations.

The advantage of this type of method lies in its simplicity and efficiency, enabling rapid prediction of electromagnetic responses within a very short time. However, there are clear limitations in the

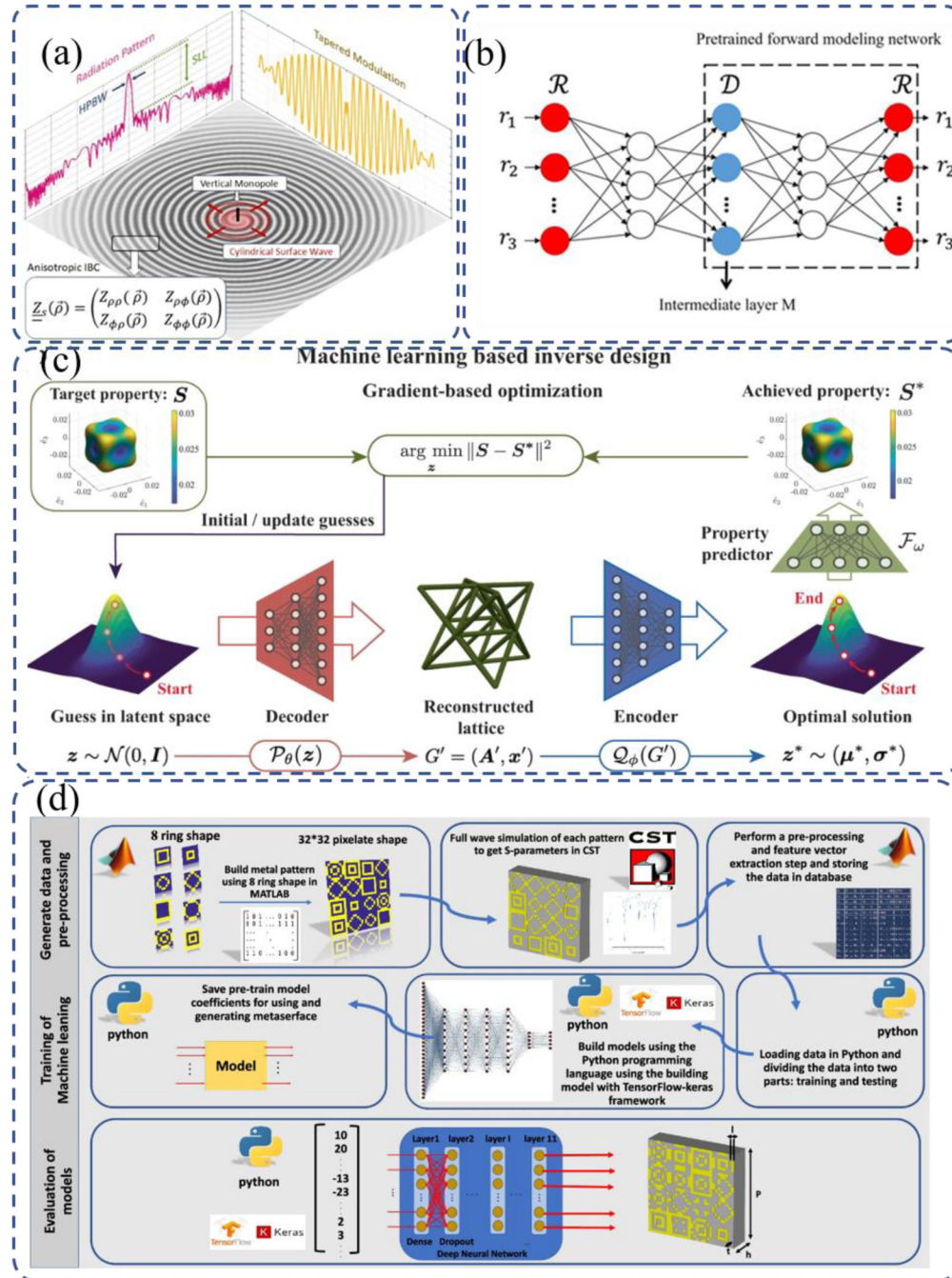
inverse design stage: on the one hand, the mapping from response to structure is many-to-one, meaning that different structures may generate similar electromagnetic functions, so relying solely on forward surrogates for searching is often inefficient; on the other hand, as the complexity of the target design increases, the search space expands exponentially, which can easily cause the design to fall into local optima. These drawbacks laid the foundation for the emergence of more sophisticated AI frameworks in subsequent research.

In 2025, Amini *et al.*¹⁷⁶ proposed a method that combines multilayer perceptrons (MLPs) with SHAP interpretability analysis to address the inverse design problem of metasurface unit structures. They first employed an MLP to establish the mapping between geometric parameters and electromagnetic responses, enabling rapid prediction of metasurface reflective unit performance and thereby significantly reducing reliance on traditional full-wave simulations, as shown in Fig. 1(a). Subsequently, the researchers introduced the SHAP method to reveal, from the perspective of explainable artificial intelligence, the influence of different geometric parameters on key performance metrics such as sidelobe level and half-power beam width. In this way, they not only achieved efficient modeling of metasurface structures but also gained a clear understanding of the specific mechanisms by which parameter variations affect design objectives.

To overcome the low efficiency of inverse mapping using forward models, researchers proposed the so-called Tandem network architecture, which represents a milestone in metasurface inverse design. The core of the Tandem architecture is to cascade a pre-trained forward prediction network with a trainable inverse network, simultaneously optimizing both components to approximate the direct mapping from functionality to structure. Liu *et al.*¹⁷⁷ subsequently introduced the “tandem” (dual-network) approach to enhance robustness. They first train a forward surrogate network as described above and then independently train an inverse generator network as a conditional generator. This design cleverly circumvents the non-uniqueness problem: as long as the generated structure can reproduce the target response through the surrogate, it is considered a feasible solution. This method reduces reliance on real structure–response labels, requiring only the data needed to train the forward surrogate, and then using the surrogate’s feedback to train the inverse network. They demonstrated that in the inverse design of pixel-level coded metasurfaces, the structures generated by the tandem model matched the simulation targets, with errors in real simulations controllably smaller than those of traditional gradient-based methods, as shown in Fig. 1(b). Nevertheless, the tandem method still depends on the accuracy of the surrogate model; if the surrogate error is large, the generated structures perform poorly in real simulations, and the method’s generative diversity remains limited.

To address the lack of diversity in the Tandem framework, generative models have been introduced into metasurface design, with representative methods including variational autoencoders (VAEs) and generative adversarial networks (GANs). These methods can learn the complex distribution between structure and functionality in the latent space, thereby generating diverse candidate solutions during the inverse process. Zheng *et al.*¹⁸⁰ developed a generative framework using a graph-based VAE combined with a

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FIG. 1. (a) illustrates the tapered modulation of surface impedance. The taper shape significantly affects the SLL and beam width; therefore, by carefully selecting the taper profile, desirable radiation characteristics can be achieved.¹⁷⁶ From Amini *et al.*, *Sci. Rep.* **15**, 24029 (2025). Copyright 2025 Author(s), licensed under a Creative Commons Attribution (CC BY) license. (b) The cascaded network is composed of an inverse design network connected with a forward modeling network.¹⁷⁷ Reproduced with permission from Liu *et al.*, *ACS Photonics* **5**(4), 1365–1369 (2018). Copyright 2018 American Chemical Society. (c) The inverse design framework aims to generate truss lattices with target properties.¹⁷⁸ From Tran *et al.*, *npj Artif. Intell.* **1**, 2 (2025). Copyright 2025 Author(s), licensed under a Creative Commons Attribution (CC BY) license. (d) A sketch of the design process for a DNN-based metasurface inverse design method. The process includes three steps: data generation and preprocessing, machine learning training, and model evaluation.¹⁷⁹ From Ghorbani *et al.*, *Sci. Rep.* **1**, 7102 (2021). Copyright 2021 Author(s), licensed under a Creative Commons Attribution (CC BY) license.

performance predictor to design truss metamaterials with customized mechanical properties. This framework supports inverse design of structures exhibiting linear and nonlinear mechanical behaviors beyond the training domain.

Tran *et al.*¹⁷⁸ and other researchers applied conditional VAEs (cVAEs) to metasurface inverse design. For example, by using spectral features as conditional inputs, they trained a cVAE to generate unit topologies or continuous geometric parameters that satisfy the conditions, as shown in Fig. 1(c). Implementation details include how to choose the latent space dimension (trade-off: too low limits expressiveness, too high increases training difficulty), how to integrate post-discretization processing (e.g., thresholding pixelated patterns), and how to use optimizers in the latent space (e.g., optimizing z in latent space to precisely match responses), thereby combining VAE with gradient-based optimization to achieve better solutions.

Wang *et al.*¹⁸¹ proposed a novel data-driven metamaterial design framework based on deep generative modeling. A VAE and a regressor for performance prediction are simultaneously trained on a large metamaterial database to map complex microstructures into a low-dimensional, continuous, and organized latent space. In this study, it was shown that the latent space of the VAE provides a distance metric to measure shape similarity, enables interpolation between microstructures, and encodes meaningful patterns of geometric and property variations.

Building on generative models, further research has begun to incorporate physical constraints to ensure the interpretability and manufacturability of the generated results. The most representative approach in this regard is the physics-informed neural network (PINN), which directly embeds Maxwell's equations or other electromagnetic knowledge into network training, allowing the model to rely not only on data but also automatically satisfy physical laws. Although the PINN framework proposed by Raissi *et al.*¹⁸² was initially applied in fluid mechanics, it was soon extended to metasurface inverse design. In metasurface research, Xu *et al.*¹⁸³ applied the physics embedding technique to the inverse design of multi-bit phase control in programmable metasurfaces and experimentally validated it, significantly improving the reliability of inverse design results. Meanwhile, Ghorbani *et al.*¹⁷⁹ proposed a model combining MLP with SHAP interpretability to reveal the causal relationship between geometric parameters and spectral responses, which is important for transparent design and engineering guidance, as shown in Fig. 1(d). These physics-constrained methods introduce a new direction for AI-based design, shifting from black-box prediction to results that better conform to physical intuition, facilitating fabrication and verification.

As research demands have further expanded, reinforcement learning (RL) has also been introduced into the field of metasurface inverse design. Unlike the aforementioned methods, RL does not rely on fixed datasets; instead, it explores the design space for optimal solutions through a "trial-error-feedback-update" process. Hooten *et al.*¹⁸⁴ employed algorithms such as PPO and DQN to train RL agents in simulated environments, enabling them to iteratively generate or adjust structures to achieve target responses. This approach is particularly suitable for online design or dynamic control scenarios, but it suffers from low sample efficiency, typically relies on surrogate models rather than real simulations, and requires policy pruning due to the large exploration space.

Compared with static generative models, RL offers advantages in adaptability and dynamism, making it especially suitable for programmable and real-time controllable metasurfaces. However, RL entails extremely high training costs, requiring extensive simulation or experimental interactions, which remains a significant practical limitation. To address this, some researchers have proposed hybrid approaches, for example, combining RL with generative models: the generative model quickly provides candidate solutions, which are then dynamically optimized by RL, balancing efficiency and adaptability.

In the past two years, more complex and general AI models have begun to enter the metasurface design field, with the most representative being transformer-based sequence modeling methods and diffusion model-based generative approaches. At the same time, diffusion models, which have become prominent in the image generation domain in recent years, have been attempted for application in metasurface design. These emerging methods indicate that metasurface design is entering a "general AI" stage, where researchers not only focus on achieving specific objectives but also aim to construct a unified design framework capable of handling multi-task, multi-band, and multi-physics coupled problems.

Before entering higher-order methods, reinforcement learning (RL) introduced a fundamentally new perspective to metasurface design. Unlike supervised learning, which depends on large input-output pairs, RL emphasizes the "agent-environment" interaction, using reward functions to drive the model to explore optimal designs continuously. Compared with forward models and generative models, RL aligns more closely with an "autonomous decision-making" design logic, demonstrating unique advantages in complex multi-objective optimization problems. For example, Fan *et al.*¹⁸⁵ proposed a deep reinforcement learning-based metasurface design method, using the metasurface reflection coefficients as environmental feedback. The agent gradually approaches the desired electromagnetic response by tentatively modifying structural parameters to gain rewards. This approach avoids reliance on a complete database and exhibits strong adaptive capability. However, its drawback lies in the extremely high training cost, particularly when the search space is highly complex, requiring millions of interaction iterations.

With the rise of reinforcement learning, researchers have begun exploring more complex AI architectures to address the high-dimensional challenges of metasurface inverse design. Transformer models, originally achieving great success in natural language processing, feature a core "self-attention mechanism" that is well-suited for handling metasurface structures with spatial correlations. In recent years, multiple studies have investigated the application of transformers in metasurface inverse design. Compared with CNNs, transformers excel at modeling global dependencies, which is particularly important for large-scale array effects in metasurfaces. For example, in communication applications, the overall reflection direction of an array depends not only on the response of individual units but also on interactions among surrounding units. These complex relationships are difficult to fully capture with convolutional networks, whereas the attention mechanism of transformers naturally incorporates global correlations into the modeling.

Another emerging approach is the diffusion model. This method has recently gained popularity in the field of image

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generation, with its core idea being to generate complex distributions through gradual “denoising.” When applied to metasurface inverse design, it can greatly alleviate the problem of multiple solutions. Compared with GANs, diffusion models are more stable during training and inherently support multi-solution generation, which is particularly suitable for metasurface design problems where a single target response may correspond to multiple structural solutions. The introduction of diffusion models marks a new stage in AI-driven metasurface design, emphasizing “exploration of diverse solution spaces” and offering new possibilities for future personalized and multifunctional metasurfaces.

In summary, from the earliest forward surrogate models, to the tandem architecture for inverse optimization, to the rise of generative models such as GANs and VAEs, followed by physics-constrained PINNs, and more recently reinforcement learning, transformers, and diffusion models, the application of AI in metasurface physical design has shown a clear evolutionary path. Each stage addresses the limitations of the previous one: forward models reduced simulation costs, the tandem architecture improved convergence in inverse design, generative models overcame the multi-solution problem, PINNs ensured physical plausibility through constraints, and reinforcement learning and emerging models further pushed AI from static prediction toward dynamic adaptability and diverse generation. Although challenges remain, such as high training costs, limited generalization, and incomplete modeling of large-scale array coupling effects, it is foreseeable that future metasurface design will increasingly rely on these intelligent approaches.^{186–222}

III. APPLICATIONS OF ARTIFICIAL INTELLIGENCE IN RECONFIGURABLE INTELLIGENT SURFACE (RIS) COMMUNICATION CONTROL

The efficient application of RIS in practical communication systems faces multiple challenges, including complex channel modeling, high-dimensional parameter optimization, and adaptation to dynamic environments. AI technologies, particularly deep learning and reinforcement learning methods, provide powerful tools to address these challenges due to their strong nonlinear fitting capability, adaptive learning ability, and large-scale optimization capacity.²²³ This chapter systematically reviews the key applications of AI in RIS communication control, including fundamental frameworks, core algorithms, experimental validation, as well as the challenges and future directions, revealing the technological pathways and application prospects of AI-RIS integration.

A. AI-based fundamental framework for RIS communication control

The core objective of an RIS communication control system is to establish an intelligent closed-loop mechanism that can adapt dynamically to environmental changes. Inspired by biological intelligence regulation mechanisms, the intelligent control framework of RIS generally consists of three key components: environment sensing, intelligent decision-making, and actuation control.²²⁴ In practical deployment, the environment sensing module acquires critical information such as user locations, link quality, and interference distribution through channel probing, terminal feedback, or external sensors.²²⁵ These data are then fed into the decision-

making unit, where AI algorithms analyze them and output optimal control strategies. The actuation module adjusts the electromagnetic characteristics of RIS elements in real time according to these strategies, by modulating bias voltages or mechanical structures.^{226,227} With this closed-loop mechanism, RIS can rapidly respond to changes in the external environment on the order of milliseconds, effectively enhancing system robustness.

In RIS control, AI technologies can be broadly categorized into two types: data-driven methods and hybrid methods that combine physical models with data-driven approaches. The former relies on large-scale data to train deep neural networks, directly establishing a mapping between inputs and RIS control parameters, offering simplicity and high adaptability.²²⁸ However, this approach typically requires a substantial amount of labeled data and lacks physical interpretability.

To address the limitations of purely data-driven approaches, researchers have proposed hybrid design schemes that combine physical models with AI.²²⁹ These methods incorporate prior knowledge such as electromagnetic theory and communication principles into network design and employ techniques like physics-constrained loss functions, structured models, or knowledge distillation to enhance both data efficiency and physical consistency.²³⁰ For instance, in RIS channel modeling, the angular sparsity of electromagnetic wave propagation and path loss patterns can guide neural network architectures;^{231,232} in beamforming optimization tasks, the principle of electromagnetic reciprocity can be embedded into reinforcement learning reward functions.²³³

Existing studies have demonstrated the significant advantages of this hybrid approach. Wu *et al.* proposed a physics-inspired holographic metasurface design network that embeds electromagnetic propagation constraints, effectively improving inverse design accuracy while reducing the need for training samples.²³⁴ Meanwhile, You *et al.* incorporated the Kramers–Kronig relations into deep learning models, as shown in Fig. 2, significantly enhancing the physical plausibility of metasurface designs.²³⁵ Therefore, the deep integration of physical laws with data-driven approaches has gradually become the main development trend in RIS intelligent control research.

B. Applications of deep learning in RIS channel modeling and optimization

Accurate channel state information (CSI) is a prerequisite for achieving efficient RIS control. Channels in RIS-assisted communication systems exhibit a dual cascaded nature (from transmitter to RIS, and RIS to receiver) and high dimensionality, posing significant challenges for conventional channel estimation methods.^{236,237} Deep learning techniques, through end-to-end learning, can effectively capture the complex characteristics of RIS channels, significantly improving estimation accuracy and efficiency.

For RIS-assisted millimeter-wave MIMO systems, Lin *et al.* proposed a sparsity-based deep learning channel estimation algorithm.²³² This method leverages the angular sparsity of millimeter-wave channels and designs a hybrid model combining compressed sensing and convolutional neural networks, achieving high-precision channel estimation while greatly reducing pilot overhead. Simulation results show that, compared to the traditional OMP algorithm, this approach

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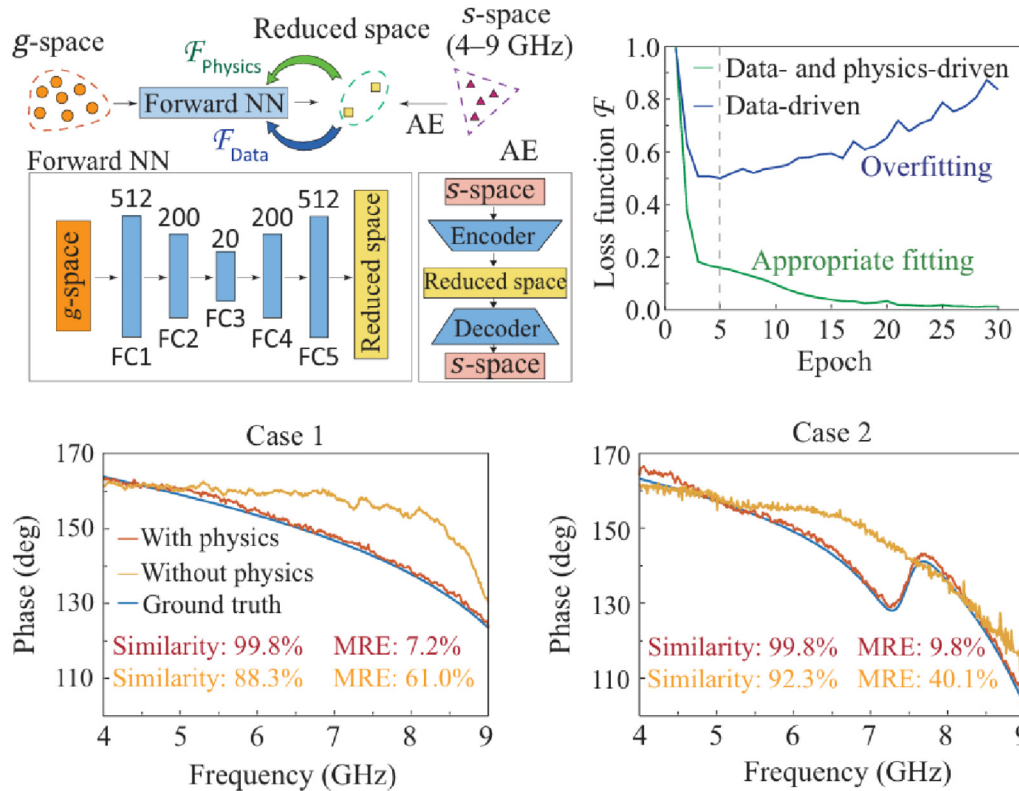


FIG. 2. Forward prediction based on the KK-assisted neural network.²³⁵ Reproduced with permission from You *et al.*, Phys. Rev. Appl. **22**, L041002 (2024). Copyright 2024 American Physical Society.

improves the mean squared error performance by approximately 3 dB under low signal-to-noise ratio conditions.

Considering the dynamic nature of practical communication environments, He and Yuan *et al.* proposed an RIS channel prediction method based on recurrent neural networks (RNNs).²³⁷ By learning the temporal evolution of the channel, this method can perform short-term predictions of future channel states, providing advanced information for dynamic RIS adjustment. Experimental results demonstrate that this method can achieve effective channel prediction at the 100 ms scale in typical indoor environments, reducing performance loss in dynamic RIS control by over 20%.

For large-scale RIS scenarios, Shukla *et al.* proposed a cascaded deep learning framework.²³⁸ This framework decomposes the high-dimensional RIS channel estimation problem into multiple subproblems, using dedicated neural network modules to separately process the transmitter-to-RIS and RIS-to-receiver links, significantly reducing computational complexity and improving estimation accuracy. In an RIS system with 1024 elements, this method achieves estimation performance close to the theoretical upper bound with ideal CSI.

RIS beamforming is a core technique for signal enhancement and interference suppression. It essentially optimizes the phase shifts of each element to coherently combine signals in the desired

direction while suppressing signals in interference directions. Traditional optimization methods face high computational complexity and poor real-time performance, whereas deep learning provides a new pathway for achieving efficient beamforming.

Considering the need for interference suppression in multi-user scenarios, del Hougne *et al.*²³⁹ designed a deep learning-based RIS multi-user beamforming algorithm. This algorithm employs a multi-task neural network to simultaneously optimize the RIS phase configuration and the base station's transmission weights, achieving interference isolation among multiple users. In office environment experiments, this method increased the number of effective MIMO channels by 2.9 times, significantly enhancing system capacity.

For RIS-assisted secrecy communication scenarios, Yang *et al.*²⁴⁰ proposed a joint beamforming optimization method based on deep reinforcement learning. This approach incorporates physical-layer security metrics into the reinforcement learning reward function, enabling the joint optimization of base station beamforming and RIS phase configuration through learning. It effectively enhances the signal for intended users while suppressing the reception quality of potential eavesdroppers. Simulation results show that this method maintains high secrecy rates under various channel conditions, achieving over a 30% improvement compared to traditional approaches.

The inverse design problem of RIS refers to determining the physical structure or control parameters of the RIS based on the desired electromagnetic response (e.g., beam direction and gain distribution), and it is a core technology for achieving customized RIS functionality. This problem faces challenges of high dimensionality, non-convexity, and multiple solutions, making traditional optimization methods inefficient.²⁴¹ Deep learning significantly improves the efficiency and flexibility of inverse design by constructing a nonlinear mapping from target responses to RIS parameters.

Liu *et al.*¹⁷⁷ were the first to apply deep neural networks to the inverse design of metasurfaces, proposing an end-to-end design framework. This framework trains a multilayer perceptron to directly learn the mapping between metasurface structures and scattering characteristics, enabling rapid design of nanophotonic structures. Compared with traditional trial-and-error methods, the design efficiency increased by more than two orders of magnitude.

To address the multiple-solution issue in inverse design (i.e., different RIS configurations may produce similar electromagnetic responses), researchers have proposed various solutions. Zhen *et al.*²⁴² introduced the tandem neural network method, which trains a forward prediction network and an inverse design network in series to alleviate the training difficulties caused by multiple solutions, as shown in Fig. 3(a). This method achieved good results in the design of transmissive metasurface cloaks, reducing design errors by 40%.

Liu *et al.*²⁴³ utilized Deep Convolutional Generative Adversarial Networks (DCGANs) for the inverse design of quad-channel full-space metasurfaces, enabling the generation of multiple metasurface structures that satisfy quad-channel full-space electromagnetic response requirements, as shown in Fig. 3(b). This approach provides researchers with abundant design alternatives, facilitating the balance of multi-channel performance, fabrication feasibility, and cost control. In the inverse design of full-space metasurfaces, the number of valid designs generated by this DCGAN-assisted method is more than six times that of traditional optimization methods while ensuring comparable quad-channel response consistency and spatial coverage performance.

Unni *et al.*²⁴⁴ proposed a deep convolutional mixture density network, modeling the outputs of metasurface inverse design as a multimodal probability distribution, which can provide multiple high-quality design options in a single inference. This method demonstrates excellent performance in multilayer photonic structure design, with both design efficiency and diversity surpassing traditional approaches.

C. Application of reinforcement learning in dynamic RIS communication control

Wireless environments in RIS communication systems are often dynamically changing, such as user mobility, obstacle blockage, and interference variation, requiring the RIS to adjust its configuration in real time to maintain communication performance. RL, by continuously interacting with the environment to learn optimal decision policies, is highly suitable for addressing such dynamic control problems.

Sajedian *et al.*²⁴⁵ first applied deep reinforcement learning to real-time RIS control, proposing a holographic optimization method for metasurfaces based on Double Deep Q-Networks

(DDQNs). This method learns the mapping between metasurface unit states and holographic image quality, enabling dynamic optimization of holographic quality in changing environments, as shown in Fig. 4(a). Experimental results demonstrate that the method can rapidly adjust to environmental changes while maintaining high holographic reconstruction quality.

For the RIS-assisted multiple-input single-output (MISO) communication system, Feng *et al.*²³³ proposed a deep reinforcement learning-based phase shift optimization method. In this approach, the RIS phase configuration is modeled as a discrete action space, and the reinforcement learning agent interacts with the communication environment to learn the optimal phase adjustment strategy under different channel conditions. Simulation results demonstrate that this method achieves a convergence speed three times faster than conventional optimization methods in dynamic channel environments, while also adapting more effectively to channel variations.

Considering the action space explosion caused by the large number of RIS elements, researchers have proposed various solutions. Wang *et al.*²⁴⁸ developed a hierarchical reinforcement learning-based RIS control method, which decomposes the high-dimensional control problem into multiple low-dimensional sub-problems. This approach enabled efficient control in an RIS system with 1000 elements, reducing decision-making time from the second level to the millisecond level, while maintaining good performance.

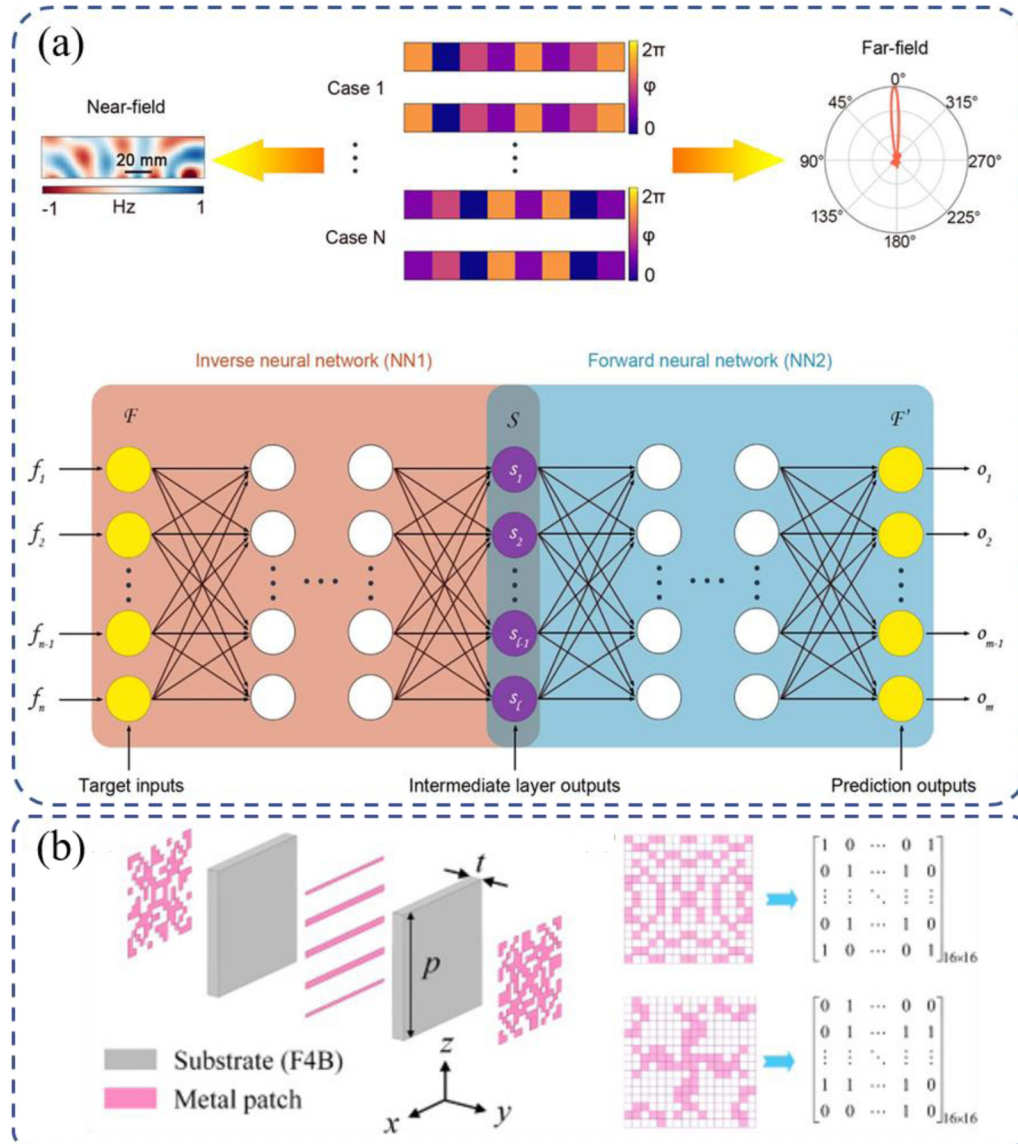
Interference and uncertainty in wireless communication environments are critical factors that affect the effectiveness of RIS control. Benefiting from its intrinsic exploration-exploitation mechanism, reinforcement learning can learn highly robust control policies, thereby effectively coping with diverse interference and environmental uncertainties.

Xiao *et al.*²⁴⁹ applied deep reinforcement learning to optimize anti-eavesdropping in RIS-assisted visible light communication systems. The proposed method learns the optimal reflection strategy of RIS to enhance the signal strength of legitimate users while generating interference to potential eavesdroppers. Compared with traditional methods, this approach maintained a higher secrecy capacity even when the eavesdroppers were mobile, achieving a performance improvement of approximately 25%.

To address the common blockage problem in millimeter-wave communications, Li *et al.*²⁴⁶ proposed an RIS control method that integrates computer vision with reinforcement learning, as illustrated in Fig. 4(b). In this method, visual sensors are used to detect changes in the positions of obstacles in real time, and reinforcement learning dynamically adjusts the RIS configuration to bypass the obstacles and maintain the communication link. Experimental results show that in dynamically changing blockage environments, the communication outage probability was reduced by more than 60%.

To enhance the robustness of RIS control against model uncertainties, researchers have proposed a variety of robust reinforcement learning approaches. For example, Bayesian reinforcement learning can be used to explicitly model uncertainties in the channel, while adversarial reinforcement learning improves the resilience of the learned policy against external interference. These methods have demonstrated superior robustness in practical tests; even with a 30% modeling error in the channel, the performance degradation can be limited to within 10%.²⁴⁰

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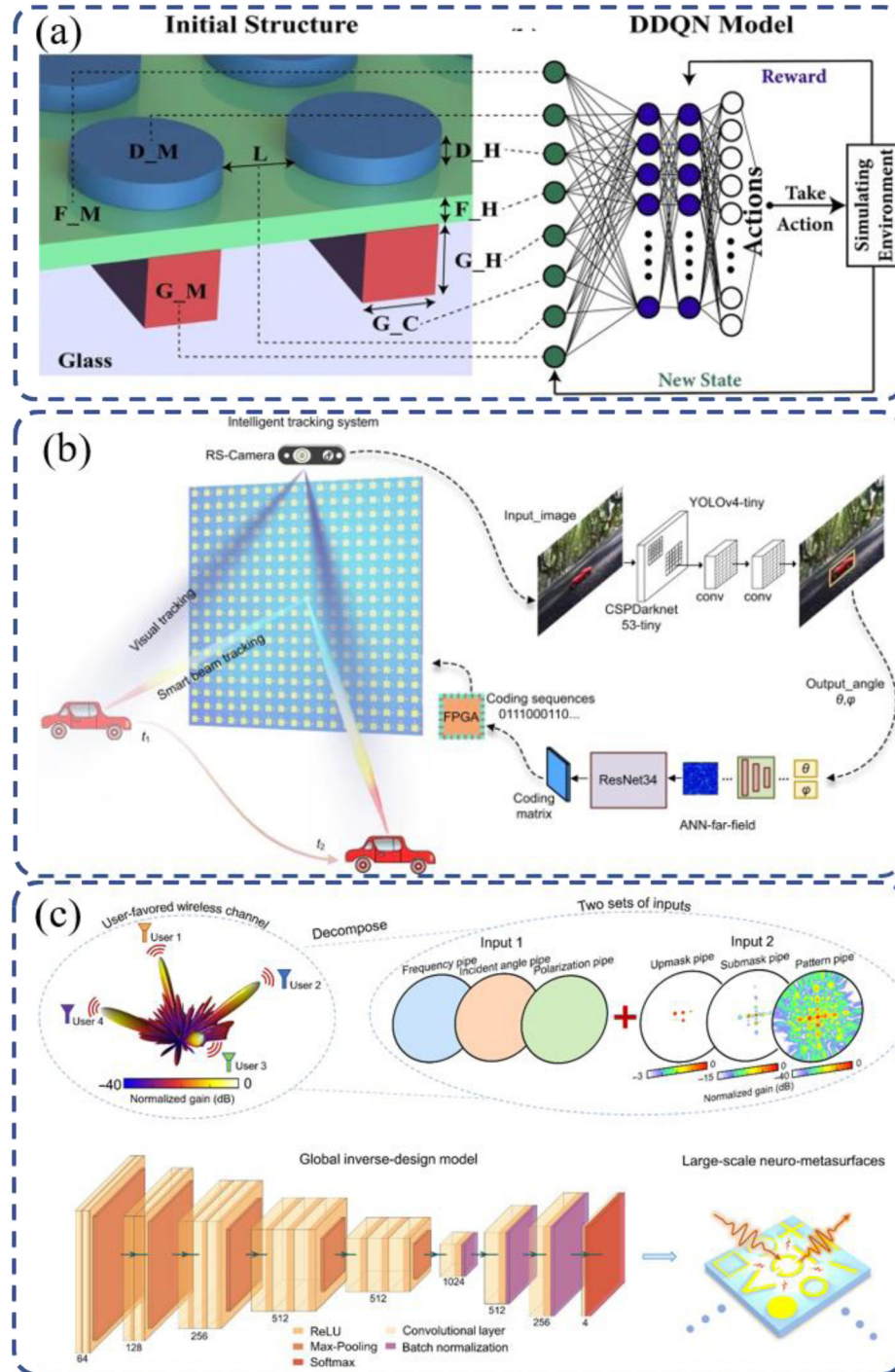
FIG. 3. (a) The T-NN in inverse design addresses the non-uniqueness problem.²⁴² From Zhen *et al.*, *Photon. Res.* **9**, B229–B235 (2021). Copyright 2021 Author(s), licensed under a Creative Commons Attribution (CC BY) license. (b) The workflow of the DCGAN-based architecture for inverse design of MSs.²⁴³ From Liu *et al.*, *Opt. Express* **32**, 46542–46556 (2024). Copyright 2024 Author(s), licensed under a Creative Commons Attribution (CC BY) license.

RIS communication control systems often require the simultaneous optimization of multiple objectives, such as system capacity, energy consumption, and user fairness, which typically involve inherent trade-offs. Reinforcement learning, especially multi-objective reinforcement learning, provides an effective framework for addressing such complex optimization problems.

For RIS-assisted multi-user communication systems, researchers have proposed a resource allocation method based on Multi-objective Deep Deterministic Policy Gradient (MODDPG).

This approach jointly optimizes the system's total throughput and user fairness, and by learning Pareto-optimal strategies, it enables flexible performance trade-offs across diverse communication scenarios.²⁵³ Simulation results show that this method reduces the variance of user data rates by 40% while maintaining the total throughput, thereby significantly improving fairness.

Considering the energy consumption of RIS itself, Wang *et al.*²⁴⁸ proposed an energy-aware reinforcement learning control strategy. This approach minimizes RIS adjustment power



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FIG. 4. (a) Illustration of the application of the DDQN algorithm in finding efficient holograms.²⁴⁵ From Sajedian *et al.*, *Sci. Rep.* **9**, 10899 (2019). Copyright 2019 Author(s), licensed under a Creative Commons Attribution (CC BY) license. (b) Integration of computer vision and convolutional neural networks (CNNs) for automatic detection of moving target positions.²⁴⁶ From Li *et al.*, *Nat. Commun.* **14**, 989 (2023). Copyright 2023 Author(s), licensed under a Creative Commons Attribution (CC BY) license. (c) Global inverse design of steady-state neural metasurfaces.²⁴⁷ From Fan *et al.*, *Sci. Adv.* **8**, eabn7905 (2022). Copyright 2023 Author(s), licensed under a Creative Commons Attribution (CC BY) license.

consumption while optimizing communication performance. By incorporating an energy penalty term into the reward function, the agent is guided to learn energy-efficient control strategies. Experimental results demonstrate that, while maintaining communication quality, this method reduces RIS energy consumption by 35%.

In scenarios where RIS collaborates with conventional base stations, Fan *et al.*²⁴⁷ developed a reinforcement learning-based joint resource allocation method. This approach simultaneously optimizes the base station's power allocation, RIS phase configuration, and user scheduling strategy to enhance overall system performance, as illustrated in Fig. 4(c). Experimental results indicate that, compared with separate optimization, the joint optimization approach improves system spectral efficiency by 20% and reduces energy consumption by 25%.

D. Application of transfer learning in RIS

In RIS communication systems, the channel characteristics and optimization objectives vary significantly across different scenarios (e.g., indoor, outdoor, urban, and rural). Training an AI model separately for each scenario would result in enormous computational and data overhead. Transfer learning effectively reduces the need for labeled data in new scenarios and accelerates the model adaptation process by transferring knowledge learned from one scenario (the source domain) to another (the target domain).

Zhu *et al.*²⁵⁰ proposed a rapid metasurface design method based on transfer learning, where knowledge learned from models trained on simple structures is transferred to the design of complex structures, thereby significantly reducing design time and data requirements, as shown in Fig. 5(a). In terahertz metasurface design, this method shortened the design time for new structures

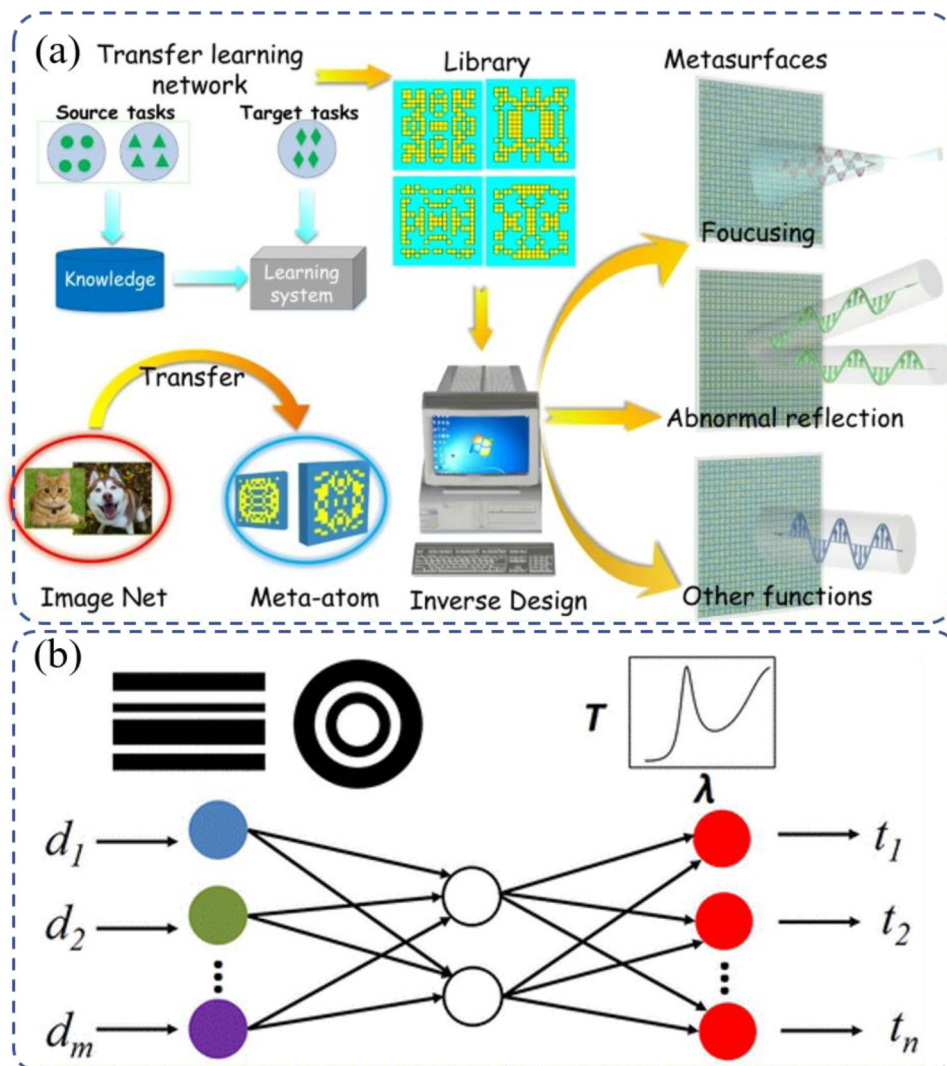


FIG. 5. (a) Illustration of metasurface design using transfer learning.²⁵⁰ From Zhu *et al.*, Nat. Commun. **12**, 2974 (2021). Copyright 2021 Author(s), licensed under a Creative Commons Attribution (CC BY) license. (b) Model based on the semi-supervised transfer learning method.²⁵¹ Reproduced with permission from Qu *et al.*, ACS Photonics **6**(5), 1168–1174 (2019). Copyright 2019 American Chemical Society.

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from several hours to the minute level, while reducing training data requirements by 60%.

For the problem of scenario adaptation in RIS channel estimation models, Fan *et al.*¹⁸⁵ proposed a transfer learning-assisted channel estimation method. In this approach, a base model trained in a typical environment is fine-tuned to adapt to a new communication environment, requiring only 30% of labeled data in the new environment to achieve performance close to that of a fully retrained model. Field tests demonstrated that this method achieved excellent transferability across different scenarios such as offices, conference rooms, and corridors, with the estimation error increasing by no more than 15%.

To enable cross-frequency transfer of RIS control strategies, Zhang *et al.*²⁵² proposed a heterogeneous transfer learning method. By designing a shared feature extractor and task-specific classifiers, the method achieved knowledge transfer from the Sub-6 GHz band to the millimeter-wave band, addressing the issue of scarce data in the millimeter-wave band. Experimental results showed that this transfer method improved the beamforming performance of millimeter-wave RIS systems by 25%, reaching 85% of the performance level achievable with sufficient data.

The scalability and functional diversity of RIS structures require AI models to possess strong modularity and reusability. Synthetic neural network and modular neural network approaches achieve efficient reuse and combination of RIS design knowledge by constructing reusable “knowledge modules.”

Lin *et al.*²⁵³ applied this concept to the assembly optimization of reconfigurable intelligent surfaces, proposing an RIS configuration method based on synthetic neural networks. This approach associates different functional RIS modules with corresponding neural network modules, achieving efficient optimization of large-scale RIS systems through the physical assembly of RIS modules and the algorithmic composition of neural network modules. In an RIS system with 10 000 elements, this method achieved an optimization efficiency two orders of magnitude higher than traditional approaches.

To enable modular reuse of RIS control strategies, researchers proposed a skill transfer method based on reinforcement learning. This approach decomposes RIS control tasks into multiple basic skills (such as beam alignment, interference suppression, and energy control), each implemented by a dedicated sub-policy, and handles complex scenarios through the combination of these skills.²⁴⁰ This method demonstrated excellent adaptability in dynamic environments, improving system response speed by 40%.

Training AI models for RIS communication systems typically requires large amounts of labeled data, which is costly to obtain in practice. Transfer learning, semi-supervised learning, and self-supervised learning provide effective means to improve data efficiency, significantly reducing dependence on labeled data.

To address the data scarcity problem in RIS inverse design, Qu *et al.*²⁵¹ proposed a semi-supervised transfer learning-based design method, as shown in Fig. 5(b). This method leverages a large amount of unlabeled metasurface scattering data along with a small set of labeled data for training, and by designing an appropriate consistency loss function, it fully exploits the information contained in the unlabeled data. Experimental results showed that the method maintained over 90% design accuracy even when the amount of labeled data was reduced by 70%.

To enhance sample efficiency in RIS channel estimation, researchers proposed a self-supervised pretraining approach. This method designs pretext tasks (such as channel map completion and noise removal) to pretrain the model using unlabeled channel data and then fine-tunes it with a small amount of labeled data, significantly improving data efficiency.²³² In millimeter-wave RIS systems, this approach reduced the required labeled data by 60% while improving estimation performance by 10%.

The current research landscape in AI and deep learning (DL) is defined by a pivotal transition from theoretical exploration and brute-force model scaling to a refined focus on practical utility, algorithmic efficiency, and seamless integration into physical systems.^{221–234} This evolution is underpinned by several critical advancements. First, the “efficiency revolution” has become a central theme, moving beyond simply increasing parameter counts in large language models (LLMs) and vision transformers (ViTs). Techniques such as unsupervised reinforcement learning frameworks, neural network pruning, quantization, and sophisticated knowledge distillation methods are dramatically reducing computational overhead and energy consumption by up to 80%, thereby democratizing access to powerful AI by enabling deployment on edge devices and embedded systems. Second, the field is rapidly advancing toward multi-modal and embodied AI, where systems are no longer siloed to process text or images in isolation but are designed to perceive, reason, and interact with the world through a fusion of sensory inputs—vision, audio, tactile feedback, and LiDAR—as exemplified by autonomous robots performing complex, dexterous tasks with over 99% accuracy. This is closely linked with the rise of digital twin technology, where AI agents can undergo millions of training cycles in photorealistic simulated environments before transferring their learned policies to the physical world. Third, algorithmic innovations in multi-task learning (MTL), such as the development of the Multi-objective Adaptive Nesterov-like acceleration (MO-AdaNL) algorithm, are effectively addressing perennial challenges like gradient conflict and negative transfer, enabling a single model to efficiently master multiple related tasks, such as simultaneous scene parsing and depth estimation, with significantly improved convergence and performance. Furthermore, the nascent integration of quantum computing through quantum machine learning (QML) is beginning to offer exponential speedups for specific problem classes, such as molecular simulation for drug discovery and complex optimization, heralding a new era of computational capability. The research prospects across various domains are consequently vast: in healthcare, AI is poised to revolutionize personalized medicine through real-time analysis of multimodal patient data; in industrial sectors, AI-driven predictive maintenance and fully autonomous supply chains will become the norm; in scientific discovery, AI will act as a co-pilot for generating hypotheses, designing experiments, and analyzing results across fields from material science to cosmology, fundamentally accelerating the pace of innovation.^{235–249}

The application of these advanced computer algorithms, particularly deep learning, has fundamentally transformed the design, optimization, and discovery of surface micro-nano and metastructures. Traditionally, the design of these structures—which manipulate light, sound, and other waves at sub-wavelength scales to achieve extraordinary functionalities—relied heavily on

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computationally expensive, iterative numerical simulations (e.g., Finite-Difference Time-Domain, FDTD) and intuitive human-driven parameter sweeps, which were prohibitively slow and limited to exploring a narrow fraction of the possible design space.^{250–253} The integration of AI has inverted this paradigm, shifting it to a data-driven, inverse-design framework. Deep learning models, especially CNNs and U-Net architectures, are now routinely used as powerful surrogate models that learn the highly non-linear mapping between a nanostructure's geometry (e.g., the shape, size, and arrangement of meta-atoms) and its resultant electromagnetic or acoustic response. This allows for near-instantaneous prediction of scattering spectra, field profiles, and phase distributions, bypassing the need for massive supercomputing clusters. More significantly, generative AI models, such as generative adversarial networks (GANs) and variational autoencoders (VAEs), can perform inverse design: given a desired optical function—such as a specific hologram, a complex beam-shaping profile, or a target absorption spectrum—the AI can generate entirely novel, often counter-intuitive, nanostructure geometries that fulfill that function with high efficiency. This capability has been pivotal in designing complex multi-functional metasurfaces for applications in imaging, spectroscopy, and LiDAR. A critical innovation enhancing the robustness and physical plausibility of these AI-generated designs is the emergence of Physics-Informed Neural Networks (PINNs), which incorporate fundamental physical constraints (e.g., Maxwell's equations) directly into the model's loss function, ensuring that the outputs are not just statistically plausible but also adhere to the laws of physics. This synergy has led to record-breaking demonstrations, such as the AI-driven design of a single metasurface that can generate over 100 independent holographic images by multiplexing various light properties (wavelength, polarization, and angle) as encryption keys, a feat virtually impossible through conventional design methodologies. The research progress thus signifies a move from designing simple, passive structures to intelligently orchestrating complex, multi-functional, and dynamic meta-devices.

Looking forward, the research prospects for AI in the domain of metasurface-based communications, or “metasurface-driven wireless communications,” are exceptionally promising and poised to form a cornerstone of next-generation (6G and beyond) networks. This field, centered on RIS or Large Intelligent Surfaces (LIS), envisions turning entire indoor and outdoor environments into programmable, intelligent transceivers that can dynamically control electromagnetic waves. AI will be the indispensable brain that controls this smart electromagnetic skin. The immediate research progress involves using deep reinforcement learning (DRL) and advanced beamforming algorithms to dynamically configure the phase shifts of thousands of individual meta-atoms on a surface in real time. This allows the RIS to perform exquisite wave manipulation: shaping and focusing beams toward intended users, creating constructive interference in signal-blind spots, and actively suppressing interference from unwanted sources, thereby dramatically enhancing energy efficiency and network capacity. The future prospects, however, extend far beyond simple beam steering. We are moving toward the development of end-to-end AI-native communication systems where the RIS, controlled by a central AI orchestrator, becomes an active part of the data transmission

process itself. This involves implementing complex mathematical operations on incoming waves, such as spatial differentiation and analog Fourier transforms, effectively turning the metasurface into an ultra-low-power, ultra-low-latency analog computer that can encode and decode information directly in the wireless domain. This concept of “over-the-air computation” could revolutionize applications like massive IoT sensor network aggregation and federated learning at the edge. Furthermore, AI will enable joint optimization of the entire transmission chain, from the base station's digital signal to the analog configuration of the RIS and the receiver's processing, creating a seamless and adaptive “smart radio environment.” The most profound long-term prospect lies in the convergence of metasurface-based communication with joint sensing and imaging. An AI-controlled RIS could simultaneously function as a high-resolution radar or hyperspectral imager, sensing the number, location, and movement of users and objects in its environment, and then using this perception to continuously and autonomously reconfigure the communication channels for optimal performance. This will be critical for enabling the high-precision localization and environmental awareness required for transformative applications like autonomous vehicle swarms, collaborative robotics in smart factories, and pervasive augmented reality. In essence, AI will not just optimize communications through surfaces; it will transmute the surfaces themselves into intelligent, sensing, and communicating entities, creating a truly ubiquitous and cognitive wireless ecosystem.

IV. SUMMARY AND OUTLOOK

In simple terms, RIS is a technology that can flexibly manipulate wireless signals, and artificial intelligence has become a key enabler for maximizing its potential, providing critical support in both “structural design” and “practical communication applications.” In RIS structural design, creating an RIS that meets specific requirements traditionally relied on repeated trial-and-error, which was time-consuming and heavily dependent on experimental data. AI has changed this paradigm: it can quickly infer the optimal RIS structure based on the desired signal response, eliminating the need for conventional trial-and-error approaches. Moreover, AI can integrate physical principles to prevent the design of impractical structures, while reducing reliance on experimental data, making RIS design more efficient and flexible.

In the communication domain, AI helps RIS address many practical challenges. For instance, as signals are transmitted, they are affected by environmental factors such as wall obstructions or user mobility. AI can accurately capture these changes, enabling RIS to adjust signal direction and strength in real time, ensuring communication quality. In multi-user scenarios, AI can coordinate RIS parameters to balance the needs of different users, preventing situations where some users experience a poor signal while others have a strong signal. Even in cases involving interference or privacy protection, AI allows RIS to adaptively manage signals to reduce interference and safeguard information security.

Currently, AI has helped RIS transition from a laboratory concept to practical applications, such as improving indoor signal coverage and enhancing the stability of high-speed communications. However, there remain areas for improvement, such as

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simplifying AI models for better compatibility with real hardware and ensuring that AI fully respects physical signal propagation laws to guarantee practical RIS adjustments.

Looking ahead, as AI technologies continue to advance, RIS will play an increasingly significant role in diverse communication scenarios, enabling seamless indoor coverage, supporting faster wireless transmission, and integrating with other technologies to become a cornerstone of next-generation wireless networks, providing more stable and efficient communication.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Haojie Zhan: Writing – original draft (equal). **Huizhen Feng:** Writing – review & editing (equal). **Yongxing Jin:** Writing – review & editing (equal). **Ying Tian:** Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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